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Techno-Economic Analysis of Hydrogen Transportation in Existing Natural Gas Pipeline

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Abstract

The aim of this study is to conduct techno-economic analysis for calculating the levelized cost of hydrogen transportation (LCOHT) in both an existing natural gas (NG) network and in new pipelines. Therefore, the pipeline retrofitting cost was considered. The analysis is based on the results of modeling and simulation of the NG-H₂ mixtures in the pipeline that provided the profile of the main parameters such as pressure, temperature and calorific values. Consequently, the economics analysis of repurposing an existing NG pipeline for H₂ transportation for a 20-years project period was conducted and including the retrofitting costs of pigging stations, valve stations, and measuring stations.

The investigation revealed that the levelized cost of hydrogen transportation (LCOHT) decreases from 0.83 to 0.46\$/kg-H₂ for NG-H₂ blends (10 to 50% H₂) respectively. However, establishing a new pipeline demonstrates a direct increase in LCOHT. For instance, the cost of new pipeline starts from 312 M\$ to 324 M\$ in comparison to 1.363-57.27 M\$ for transporting the equivalent NG-H₂ blend amount (up to 50% H₂) in repurposed pipeline. These results recommend repurposing the existing pipeline over the establishment of a new pipeline. The results are compared and concluded to support the oil and gas industry in deciding on the repurposing of the existing natural gas network.

Introduction

Worldwide different technologies are being developed to produce green hydrogen to shift the world to a more sustainable, green, and renewable energy. Transportation of the hydrogen is one of the key challenges in the hydrogen value chain both from the safety and cost, especially when it comes to high-capacity commercial use for long distance and periods (Lee & Lee, 2022).

Grey hydrogen, which is primarily derived from fossil fuels such as natural gas, plays a key role in the current hydrogen market. The leading production method of grey hydrogen is Steam Methane Reforming (SMR). In SMR methane reacts with high-temperature steam to produce Hydrogen (H₂) and carbon monoxide (CO) (Ji & Wang, 2021; Katebah & Linke, 2022). However, concerns arise over the environmental impacts of this technology due to the carbon emissions associated with it.

While steam methane reforming (SMR) effectively separates hydrogen atoms from methane molecules, the process also generates significant carbon emissions. The production of hydrogen results in the emission of around 11 kilograms of CO₂, which constitutes a substantial portion of the greenhouse gas effect (Ocko & Hamburg, 2022). Despite this drawback, grey hydrogen offers cost-effectiveness and readily accessible technology, establishing its dominance in the global hydrogen market, representing approximately 95% of production (Boretti & Banik, 2021).

Grey hydrogen continues to dominate hydrogen production methods; however, its reliance on fossil fuels and consequent high carbon emissions poses notable environmental challenges (Hassan et al., 2023). As the world progresses towards sustainability, the future of grey hydrogen hinges on its capacity to adopt cleaner production techniques or serve as a catalyst for the advancement of a more environmentally conscious hydrogen economy.

Blue hydrogen emerges as a pivotal approach in hydrogen production, serving as a transitional strategy towards achieving decarbonization in the hydrogen sector. Similar to grey hydrogen, blue hydrogen primarily relies on natural gas as its feedstock (Tetteh & Salehi, 2023). The production of blue hydrogen predominantly involves integrating steam methane reforming (SMR) with carbon capture and storage (CCS) technology (Ajanovic et al., 2022). This integration allows for the capture and storage of carbon dioxide emissions, setting blue hydrogen apart from conventional grey hydrogen production methods.

In the attempt of a more sustainable energy, green hydrogen holds great promise as a clean, sustainable energy source. Moreover, it is seen to play a role in the low-carbon economy transition significantly. Green hydrogen, produced through the electrolysis of water using renewable energy sources such as wind, solar, or hydroelectric power, represents a shining example of sustainability in the energy sector (Marouani et al., 2023).

This carbon-neutral production method entails electrolyzing water molecules (H₂O) to separate hydrogen (H₂) and oxygen (O₂). This process guarantees the absence of greenhouse gases (IRENA, 2021). Different electrolysis technologies, including alkaline electrolysis, proton exchange membrane (PEM) electrolysis, and solid oxide electrolysis, present varying efficiencies, scalability, and costs, thereby influencing the framework of green hydrogen production.

Hydrogen can be transported in two different forms, gaseous and liquid and can be delivered from one location to another. The hydrogen in gas phase could be transferred via pipelines similar to the pipelines used for the transportation of natural gas, whereas liquid hydrogen is transported by means of tanker trucks and ships. Hydrogen transportation methods vary according to the distance and quantity to be delivered; logistical considerations encompass pipelines, roads, and ships (Gerboni, 2016).

Trucks transport hydrogen in smaller volumes or in regions lacking pipeline infrastructure. This method accommodates both gaseous and liquid forms of hydrogen, utilizing specialized cryogenic containers for liquefied hydrogen and tube trailers for compressed hydrogen gas (Zhang et al., 2024). In addition, hydrogen can be bound chemically to a liquid carrier such as ammonia or organic compounds (LOHC). These carriers have the capacity to carry a larger volume of hydrogen compared to compressed or liquid hydrogen. But many challenges are associated with this method, such as releasing hydrogen from the carriers and maintaining its purity (Niermann et al., 2019; Preuster et al., 2017).

Like natural gas, a dedicated pipeline system can be established for transporting hydrogen. However, due to the small size of hydrogen molecules, it necessitates compatible materials and specially designed or upgraded pipeline infrastructure to mitigate embrittlement and leakage risks. Pipeline transportation is typically employed for longer distances and larger quantities of hydrogen. Since pipelines are underground, the probability of leaks, explosions, and other accidents, as well as environmental disruptions, is minimized (Ali et al., 2024; Gerboni, 2016b).

Ships are employed for hydrogen transportation as well. Tankers designed for cryogenic liquids or chemicals can accommodate both liquid hydrogen and hydrogen carrier liquids. However, this approach

is still being explored and encounters obstacles concerning safety, infrastructure, and expenses (Zhang et al., 2024).

Several applications and research in different countries around the world have conducted studies to evaluate the feasibility of transporting hydrogen within existing natural gas pipelines.

Currently, there are around 4600 km of hydrogen pipeline distributed across the United States and Europe (IEA, 2022). On the other hand, the natural gas pipeline has over 1.2 million km of infrastructure worldwide. The natural gas pipeline is seen as a fast-track path to deploy low emission gasses. Hydrogen blend in natural gas pipeline is seeing great interest in research. Some findings indicate that no major retrofitting is needed if hydrogen blend is between 5% to 10%. Another research has found that up to 20% is possible (IEA, 2022).

Different projects for hydrogen blend in natural gas pipelines are being deployed and targeted to start commissioning in the coming years. Such project can be found in Australia where the blend is initiated at a distribution level. Future projects are planned by the end of 2023 in both United Kingdom and Slovakia with a blend of 20% and 5% respectively. Additionally, South Korea is also planning to blend hydrogen by 20% in the natural gas pipeline by 2026 (IEA, 2022).

A study was conducted in London to analyse the effects of blending 4.8%-20% hydrogen into a natural gas pipeline through a T-junction, as well as the effects on the pipeline material and the fluid behaviour within the pipeline. The carbon steel pipeline studied had a diameter of 18 inches and operated at a pressure of 69 bar, with a gas temperature ranging from 5 °C to 45 °C and a speed of 10 m/s. The study results have shown that increasing the concentration of blended hydrogen will cause hydrogen embrittlement in the walls of high-tensile steel pipelines (Eames et al., 2022). The use of a plastic material pipeline, such as polyethylene pipeline, with low pressure has achieved the transportation of pure hydrogen (Northern Gas Net Works, 2016).

Safety issues were evaluated based on the hydrogen blending concentration in natural gas pipelines, with concentrations of 5%-15% showing minor issues. These issues began to increase with hydrogen concentrations of 15%-50%. Major issues affecting safety, pipeline material, and end users arose when the hydrogen concentration reached 50% and above. The results of analyzing the compression processes at two pressures, 16 bar and 84 bar, showed that increasing the concentration of blended hydrogen decreased the transmitted flow rate, leading to a high pressure drop due to the decrease in compressor discharge power (Witkowski et al., 2018).

The study conducted by (Koo et al., 2023) on hydrogen blending in natural gas pipeline highlighted pressure drop and erosional velocity. The outcome resulted in that 20% of hydrogen blended in natural gas was achieved without any concerns in the erosional velocity (Koo et al., 2023).

The German government aims to operate 9700 km of hydrogen pipelines, where 60% of the existing natural gas pipelines will be repurposed to achieve this goal (Steinberg, 2023). The Australian projects of hydrogen- natural gas blending pipelines aim to supply 710 houses. The blend will start at 5% and gradually increase to 10%. However, there is a regulatory limit for the allowable hydrogen blend in the natural gas carbon steel AS2885 pipeline, which is set at 13% due to safety concerns governed by government regulations (Arup et al., 2019).

The motivation of this research of repurposing NG pipelines for hydrogen transportation are the availability of NG pipeline network throughout Oman and societal acceptance in terms of rights of way and road use. Hydrogen transportation poses significant challenges, particularly in terms of capacity and distance. Thus, complex plans, approval procedure, and costly establishment of new pipeline construction are necessary for Hydrogen transportation. Such issues can be resolved by using the existing gas infrastructure. However, technologies for converting natural gas infrastructure to hydrogen operation is not an easy task. Special requirements, tools and a high level of expertise are needed. In consonance with Paris Agreement by limiting global warming to 1.5°C and reaching Net Zero Emission by 2050, Oman is expected to play a crucial role in Hydrogen production and exportation. Oman will be the highest Hydrogen volume exporter in the middle east by 2030. The exportation volume of hydrogen by 2050 will be approximately

double the volume of exported LNG in 2021 i.e., Hydrogen exported volume in 2050 to reach 8.5 MT [Energy Agency, 2023].

Methodology

Commercial software (Aspen Hysys, Aspen Plus, and PSI) were used in analyzing the Natural gas (NG)-H₂ blends properties. An existing NG pipeline segment within a gas network was considered for this analysis. The study used various thermodynamics equations of state such to calculate the gas mixture properties. Moreover, a spreadsheet is developed to run techno-economic analysis to calculate the levelized cost of H₂ blend (0- 50% H₂) transportation in pipeline. It covers the economic parameters, i.e. a detailed capital expenditure and operational expenditure.

Therefore, a techno-economic analysis was developed based on the commercial simulation results of Natural gas (NG)-H₂ blends properties i.e., commercial software such as (Aspen Hysys, Aspen Plus, and PSI) were used in simulating an existing NG pipeline segment within a gas network pipeline. This segment is connecting compressor station (central station) to an end user by the end of the of the pipeline segment. Consequently, the spreadsheet is created to run techno-economic analysis that calculates the levelized cost of hydrogen transportation (LCOHT) in the existing network pipeline (e.g., covering NG-H₂ blends range (0- 50% H₂). The economic study covers the economic parameters such as (CAPEX, OPEX, residual value, discount rate, tax rate, tax saving, discounted OPEX, annualized CAPEX, depreciation, etc.). Furthermore, the research runs the pipeline retrofitting cost. The study applies advanced cost estimation methods to analyze short and long-term financial viability and provides detailed insights into CAPEX, OPEX, LCC and LCOT i.e. the study includes:

- Quantification Phase: Assessing infrastructure requirements and estimating CAPEX.
- Retrofitting: Calculating the cost of necessary upgrades such as compressors and valve stations.
- Estimated Operating Cost: Estimating the ongoing operating costs including energy, maintenance, and monitoring.
- Life Cycle Cost (LCC) Analysis: Determining the total costs over the life of the infrastructure.
- LCOHT (Levelized Cost of Hydrogen Transportation): Calculating the average cost of hydrogen transportation per kilogram over the pipeline's life cycle, accounting for CAPEX, OPEX, and total hydrogen transported.
- CAPEX: Hydrogen mixing requires incremental investment, and costs increase significantly for higher hydrogen concentrations. However, gas pipeline modifications, for pure hydrogen transportation, account for 20 per cent of the costs of constructing new pipelines.
- OPEX: Operating costs increase with hydrogen concentration due to the increase in monitoring, maintenance, and energy requirements.
- LCC: Life-cycle costs highlight economies of different hydrogen concentrations
- LCOHT (Levelized Cost of Hydrogen Transportation): Efficient use of infrastructure at higher hydrogen concentrations reduces transportation costs

Economy Analysis Models

Economic Transformation Model

The economic transformation model serves as the framework for completing the analysis and is summarized in Table (1). The economic model begins with the Quantification phase, which involves evaluating the quantities required for all stages of the project and estimating the capital costs, from the hydrogen blending process to the end user. The second phase is based on evaluating these assets for their suitability of materials

and components for different hydrogen concentrations using theoretical foundations and the findings of the referenced studies. Accordingly, primary inspections, testing, and evaluations form part of the initial cost. Then, estimation of the adaptation or retrofitting costs is done. This involves developing the initial cost reduction by calculating the upgrading or modification costs for already installed units such as pipelines and stations. These calculations are based on 1 km as a reference, with the possibility of adapting them to other future projects, which represents one of the main objectives. Hence, the capital expenditures (CAPEX) are calculated, including expenses associated with establishing, developing, and improving long-term assets. Additionally, the operational expenditure (OPEX) is executed, which encompasses the costs associated with daily operation and maintenance. Finally, Total Cost of Ownership (TCO) and the 20-year expenditure for each hydrogen blend are determined as the Lifecycle Cost (LCC).

Table 1—Economic transformation model

Step		Description	Cost Components
1. Quantification		Assess the number and total length of gas infrastructure for hydrogen transport integration.	Data collection and infrastructure assessment costs.
2. Capital Expenditures (CAPEX)	2.1. Evaluation	Evaluate the suitability of assets, materials, and components for different hydrogen concentrations.	Inspection, testing, and evaluation costs.
	2.2. Adaptation Retrofitting Costs	Identify and define necessary adaptation measures to enable hydrogen transport, including equipment upgrades and modifications.	Retrofitting, equipment and replacement costs.
3. Operational Expenditures (OPEX)		Estimate the ongoing costs for operating, maintaining, and monitoring the repurposed infrastructure.	Energy usage, maintenance, monitoring, and staffing expenses.
4. Total Cost of Ownership (TCO)		Calculate the overall cost of repurposing gas infrastructure for hydrogen, incorporating both capital and operational expenses.	Summation of adaptation and operational costs.
5. Lifecycle Cost (LCC)		The total costs incurred throughout the entire lifecycle of a project or asset	

Estimation the Asset Quantities

Table 2 shows an overview of the assets considered. The following quantities were estimated for the pipeline segment:

Table 2—Overview of the Considered Assets

Infrastructure item	Specific asset volume per 1 km pipeline	Unit
Steel transmission pipelines	1	Km
Valve stations	0.067	station / km
Valve stations (needed for pure hydrogen service and H ₂ % 40-50%)	0.05	station / km
Pigging stations	0.015	station / km
Metering stations	0.0039	station / km
Compressor and Heat Exchanger stations	Calculated separately based on its size	-

Economic Input Parameters

Table 3 and 4 represent economic input parameters used for economic analysis. Pipeline economic lifetime reduced from 40 to 30 years due to hydrogen embrittlement (Repurposing Natural Gas Pipelines for Hydrogen: Limits and Options from a Case Study in Germany, 2024) (Extending the European Hydrogen Backbone, 2021) (Gaseous Inhibitors: A Comprehensive Overview on Mitigating Hydrogen Embrittlement in Pipeline Steels, 2024).

Table 3—Economic Input Parameters

Parameter	Value	Unit
Pipeline CAPEX	Retrofitting Section	\$
Pipeline OPEX	Table 2.	of Pipeline CAPEX
Cost of the Pipeline	1006.64	\$/m
Compressor OPEX	4%	of Compressor CAPEX
Inhibitor OPEX	0.067	\$/kgO2
Pipeline economic lifetime	20	years
Compressor lifetime	20	years
Electricity price	0.065	\$/KWh
Inflation	2%	%
Residual Value (RV)	5%	%
Real Discount rate	5.88%	%
Nominal Discount Rate (DR)	8%	%
Tax Rate	5%	%
End-of-Life Costs	10%	%

Table 4—Economic Input Parameters for pipeline OPEX

Hydrogen Percentage (H2%)	0	10	20	30	40	50
OPEX (%)	2%	2.3%	2.6%	2.9%	3.2%	3.5%

Capital Expenditure (Capex Total)

The Capital expenditure is the sum direct and indirect costs for each hydrogen blending concentration as shown in equation 1 below. Direct costs include pipeline retrofitting costs, valve, metering and pigging stations costs, compressors and heat exchanger costs. Indirect cost represents the expenses of site preparation, engineering, design, project contingency, and permitting, which are estimated at 5% of the capital expenditure.

$$\text{CapEx}_{\text{Total}} = \text{CapEx}_{\text{Pipeline}} + \text{CapEx}_{\text{Stations}} + \text{CapEx}_{\text{Comp+HE}} + \text{Indirect costs (5\%)} \quad (1)$$

Pipeline Evaluation and Retrofitting Costs

In-line inspection (ILI) technology is used for hydrogen concentrations above 10%. It involves inspecting the inside of pipelines during operation, while the gas is flowing, to detect defects such as corrosion and cracks that could impair pipeline performance. The estimated cost of ILI is \$25,692.2 per kilometer. If maintenance is required to repair cracks or similar defects, the estimated cost per repair is \$55,852.5.

Table 5—Cost assumptions for pipeline assets

H2 Concentration	Cost Type	Cost Details
Below 10 vol.-% H2	No major modification costs required	Minimal costs for general inspection and monitoring.
Up to 10 vol.-% H2	Documentation, Evaluation, and Studies	223.4 \$/km of pipeline for documentation and studies
10-50 vol.-% H2	ILI, Crack-like Defect Identification, and Repairs	ILI: 25692.2 \$/km, Repair: 55852.5 \$ per repair

Valve, Metering and Pigging Stations (Capex Stations)

Valve stations are used to control gas flow, isolate sections of pipeline and regulate pressure. Their functions include interrupting flow in emergencies or during maintenance and allowing controlled operation of certain sections of the pipeline. For the pipeline segment, 20.77 valve stations are required, as shown in Table 3, based on a rate of 0.067 stations per kilometer. With a 100 percent hydrogen mixture, however, the number of valve stations is reduced to 15.5, as hydrogen has a lower density and different flow properties than natural gas. Cost assumptions for valve stations:

- For hydrogen concentrations up to 10 %, no additional costs are required for upgrading the valve stations.
- For concentrations between 20 % and 30 %, 10 % of the valve stations will be replaced at an estimated cost of \$2 million.
- For concentrations between 40-50%, all valve stations are replaced, and one station is installed every 20 kilometers.

Pigging stations provide pipeline maintenance and cleaning while ensuring operational integrity. Their functions include in-line inspection, which cleans and inspects the pipeline from the inside, and detects issues like corrosion, cracks, and blockages. Cost assumptions for pigging stations:

Sealing is replaced for all hydrogen concentrations at an estimated cost of \$6,702.3 per station.

Metering station cost assumption:

For hydrogen concentrations up to 10%, process gas chromatography (PGC) will be renewed at a cost of \$55,852.5 per station.

For concentrations above 10%, the cost increases to \$167,557.5 per station, including \$5,585.3 to update 50% of the devices for volume converter calibration.

The cost assumptions for valve, metering and pigging stations are shown in table 6.

Table 6—Cost Assumptions for Station Assets

Station Type	H ₂ Concentration	Action Required	Cost Estimate
Valve Stations	Up to 10 vol.-% H ₂	Tightness test	No additional costs considered
	20 - 30 vol.-% H ₂	10% of valve stations replaced	2.07 Mil. \$ per replaced valve station
	40 - 50 vol.-% H ₂	All valve stations replaced by DBB stations (every 20 km)	2.4 Mil. \$ per valve station
Pigging Stations	All concentrations	Replacement of sealing	6702.3 \$ per pigging station
Metering Stations	Up to 10 vol.-% H ₂	Renewal of PGC	55852.5 \$ per metering station
	Above 20 - 50 vol.-% H ₂	Renewal of PGC and auxiliary system modification	167557.5 \$ per metering station
		Volume converter calibration/update (50% of devices)	5585.3 \$ per device

Results and Discussion

Hydrogen blending in existing natural gas network was modeled and simulated using mainly three different Equations of State. Therefore, the study examined the operational parameters such as pressure profile, temperature profile and energy analysis associated with varying hydrogen percentage (H₂%) in natural gas-H₂ blends, utilizing three commercial simulation software: Aspen Plus, Aspen HYSYS, and PSI. The analysis of the results using these EoS was conducted to view the EoS performance and reliability in modeling various compositions of natural gas and hydrogen. All EoS models showed similar overall trends. These findings have significant implications for the adaptation of existing natural gas infrastructure, necessitating adjustments to pipeline systems, modifications to combustion equipment, and reconsideration of safety protocols in gas transportation. Thus, the simulation evaluated the key parameters across hydrogen blends ranging from 0% to 50% H₂ (% volume). Significant results were obtained showing that the pressure drop along the pipeline decreases as the percentage of the hydrogen blend increases. This trend can be related to hydrogen's lower density in comparison with natural gas. Moreover, the temperature drop diminishes with increasing hydrogen blend percentages i.e., the temperature tends to be stabilized, reaching thermal equilibrium with the surrounding environment. Additionally, the energy analysis demonstrates a consistent trend i.e., as the hydrogen content increases, both the Higher Heating Value (HHV) and the Lower Heating Value (LHV) decrease. The results of the simulation were used in determining the flow rates, pressure, and temperature that are used to define the economic study requirements (e.g. the need for using compressors and heat exchangers). Also, the outcome of this part is highlighting the influence of H₂ blends on gas transport efficiency. The details of these modeling and simulation results are not shown here as it will be published in another paper since the scope of this publication is focusing exclusively on economic evaluation of hydrogen transportation in existing natural gas pipelines.

The economics analysis of repurposing an existing NG pipeline for H₂ transportation for a 20-years project period was conducted including the retrofitting costs of pigging stations, valve stations, and measuring stations. The below sections demonstrate the detailed of this analysis

Capital Expenditure

It is noted that with the increase in hydrogen concentration, the CAPEX rises, as shown in [figure 1](#). This was expected to be due to the addition and modification of components in the piping system to accommodate

pressure differences and mitigate hydrogen embrittlement. In this analysis, hydrogen blend concentrations from 10% to 50% were studied.

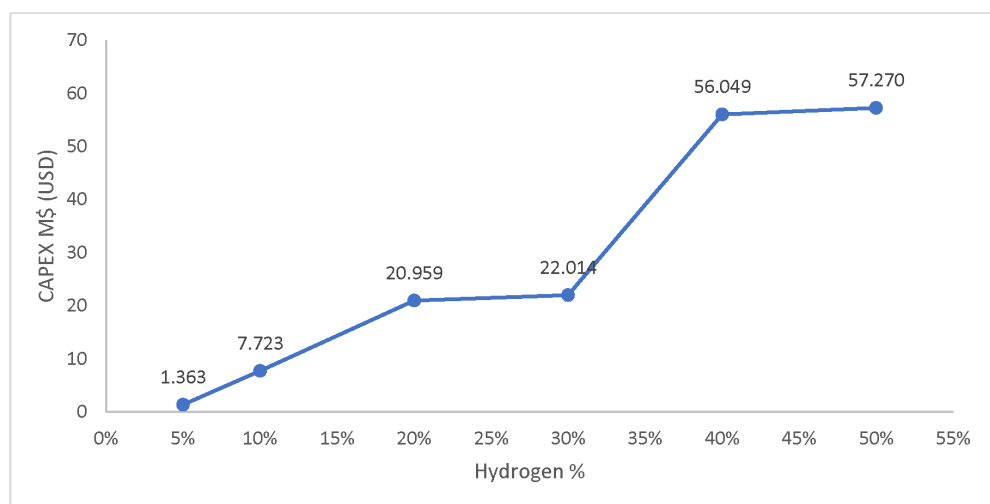


Figure 1—The relation between CAPEX and Hydrogen Concentration in NG-H2 Blend

The most significant results are as follows:

- For low hydrogen concentration (up to 10%), the cost is estimated at \$7.72 million, with compressor costs accounting for the largest share.
- For medium concentrations (20% and 30%), costs increase to \$20.96 million and \$22.01 million respectively, a difference of 5%. This increase is mainly due to the high cost of valves and compressor stations, which rise by 65% compared to concentrations below 10%.
- For high concentrations (40%–50%), the cost at 40% reaches \$56.05 million, primarily due to a significant increase in valve station costs, as they must be completely replaced to handle high pressure and hydrogen concentration.

Operation Expenditures

The OPEX costs increase as the hydrogen concentration rises as shown in [figure 2](#).

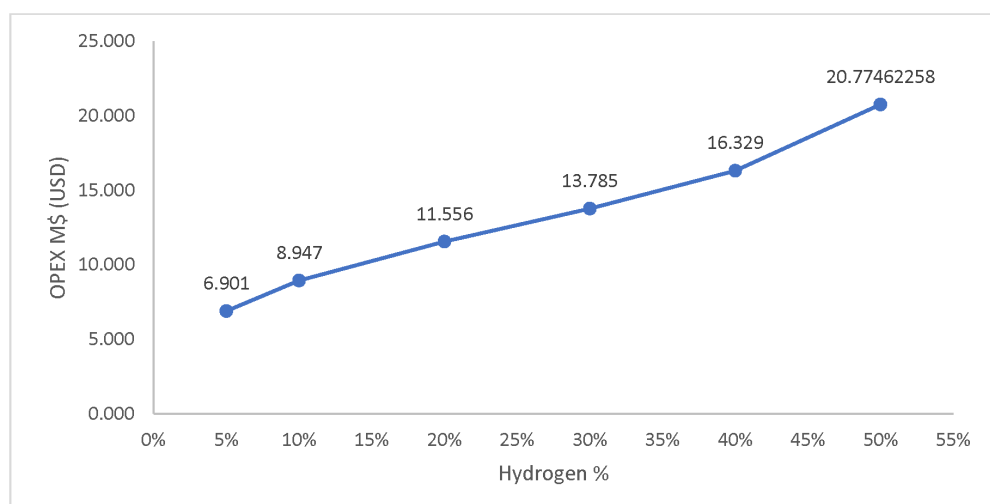


Figure 2—The relation between OPEX and Hydrogen Concentration in NG-H2 Blend

The main conclusions drawn from the study results are as follows:

- For low hydrogen concentration (10%), the operating cost is estimated at \$8.95 million, with the largest share accounted for by pipeline OPEX, costing \$7.18 million per year.
- For medium concentrations (20%–30% H₂), the operating cost increases to \$11.56 million for 20% H₂ and \$13.78 million for 30% H₂, reflecting a difference of 16%. This increase is primarily due to higher electricity costs and pipeline OPEX.
- For high hydrogen concentrations (40%–50% H₂), the operating cost for 40% H₂ is \$16.33 million, with pipeline OPEX at \$10.39 million and electricity costs at \$5.39 million per year.

Life Cycle Cost (LCC) & Levelized Cost of Hydrogen Transport (LCOHT)

As the hydrogen concentration in the pipelines increases, the economic impact is highlighted by the life cycle cost (LCC), the discounted life cycle cost (LCC) and the levelized cost of transportation (LCOHT). The results, shown in [figure 3](#) and [figure 4](#), are broken down as follows for different concentration ranges over a 20-year period:

- For a low hydrogen concentration (10%), the Life Cycle Cost (LCC) is approximately \$225.47 million, while the discounted LCC is \$117.88 million. The Levelized Cost of Transport (LCOHT) is highest at \$0.83 USD/kg hydrogen, primarily due to the inefficient use of infrastructure at this low concentration.
- For medium concentrations (20%–30% H₂), LCCs rise significantly to \$303.37 million for 20% H₂ and \$358.60 million for 30% H₂, while the discounted LCCs are \$175.79 million and \$203.33 million, respectively. LCOTs decrease to \$0.62 USD/kg for 20% H₂ and \$0.48 USD/kg for 30% H₂, reflecting improved infrastructure efficiency at these concentrations.
- For high hydrogen concentrations (40%–50% H₂), LCCs continue to rise steadily due to the significant need for infrastructure improvements, reaching \$457.78 million for 40% H₂ and peaking at \$500 million for 50% H₂. The discounted LCCs follow a similar trend, increasing from \$304.93 million for 40% H₂ to \$329 million for 50% H₂. Meanwhile, LCOHTs steadily decline from \$0.53/kg for 40% H₂ to \$0.46/kg for 50% H₂, highlighting the direct drop in LCOHT at higher hydrogen concentrations which recommend conducting future study for higher H₂ concentrations e.g., above 50%.

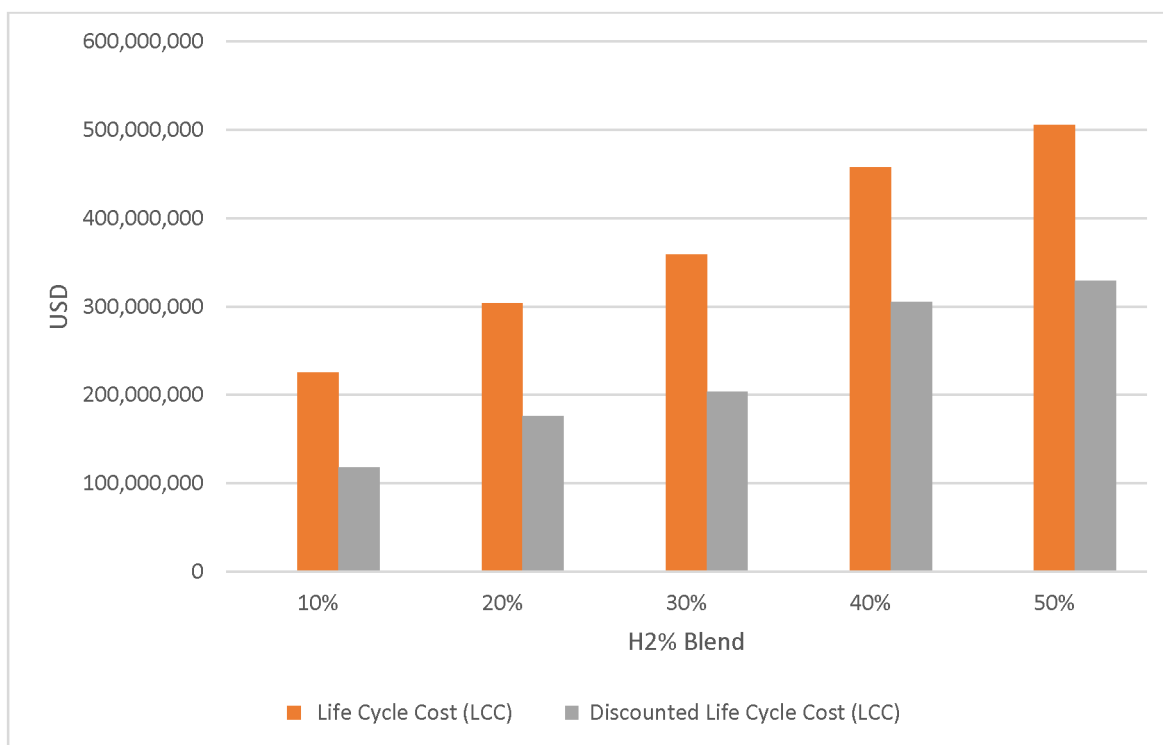


Figure 3—Life Cycle Cost (LCC) and Discounted LCC Trends for a 20-year project period

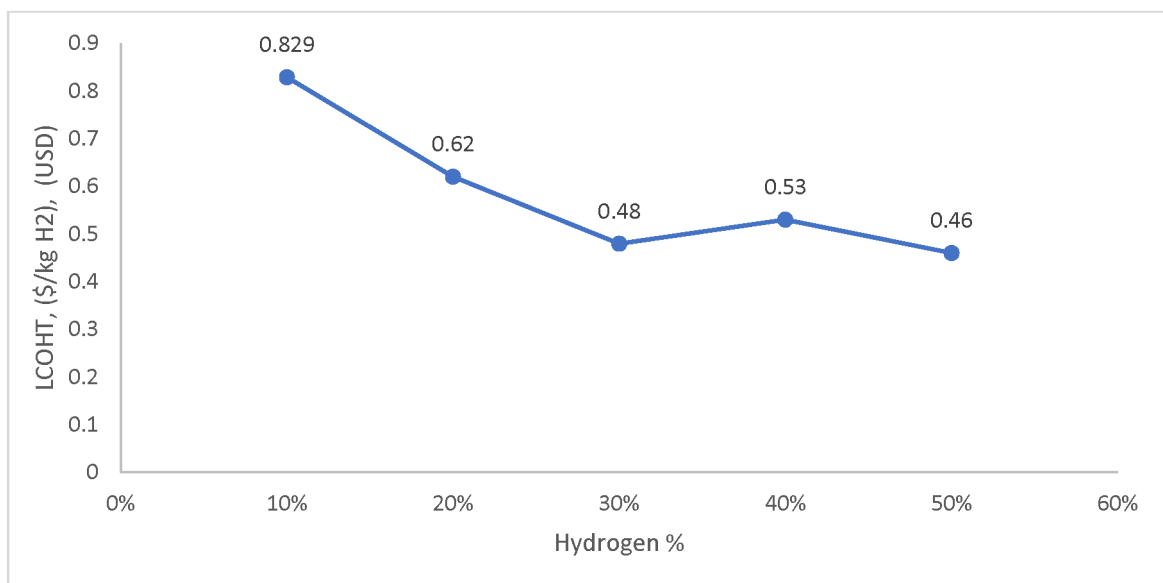


Figure 4—Levelized Cost of Hydrogen Transportation (LCOHT) for a 20-year project period

Conclusion

The investigation revealed that the levelized cost of hydrogen transportation, in existing NG network, decreases from 0.83 to 0.46\$/kg-H2 for natural gas-H2 blends (0 to 50% H2) respectively. However, establishing a new pipeline demonstrates a direct increase in LCOHT i.e., the cost of new pipeline starts from 312 M\$ to 324 M\$ in comparison to 1.363-57.27 M\$ for transporting the equivalent NG-H2 blend amount (up to 50% H2) in repurposed pipeline. These results recommend repurposing the existing pipeline over the establishment of a new pipeline (that reduces the cost up to 82%), and can be discussed due to the low retrofitting cost in comparison with high required CAPEX for establishing the new pipeline. This

finding agrees with other published results stated that repurposing a pipeline may save more than 70% of H₂ transportation cost. The findings are compared and concluded to support the Oman oil and gas industry in deciding on the repurposing of the existing natural gas network.

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